Reinforcement Learning

Assignment 1

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1 Assignment 1: Reinforcement Learning

1.1 Question 1

Q1: “Prove that the policy iteration algorithm converges to the optimal policy for a finite MDP with a finite action space for each state.“

1.2 Question 2

Q2: “Prove that the Bellman optimality equation has a unique solution for a finite MDP with a bounded reward function.“

1.3 Question 3

Q3: “Analyze the challenges of using linear function approximation in value-based reinforcement learning. Discuss the potential issues that may arise and propose methods to mitigate them“